By TOUCH DEN · November 11, 2026

Explore a Running System

Video 1 · Who, What, and Which Files

How Computers Work · Section 12

In this video, you will learn:

- How to see this whole course on a real machine
- Who you are, and what is running
- How to identify and read files

Run these on your own computer only. You are looking, not changing.

The goal: see the ideas for real

You've built a full model of a computer. This lesson makes it **real** with a few commands, the way security people start on an unfamiliar system:

- **Who am I, what can I do?** (lesson 5)
- **What is running?** (lesson 6)
- **What are these files, really?** (lesson 7)
- **How long up, how did it start?** (lessons 4, 9)

Open a terminal on **your own** computer and follow along.

Who am I, and what is running

```
$ whoami
user
$ ps -o pid,ppid,%cpu,%mem,comm
  PID  PPID %CPU %MEM COMM
 1521     1   0.3  1.8 WindowServer
 2983  2970   1.2  4.1 Google Chrome
 3001  2983   0.0  2.2 Google Chrome Helper
$ ps -e | wc -l
     900
```

- `whoami` : your account, your powers depend on it (lesson 5).
- `ps` : **PID**, **parent**, resources (Helper's parent is Chrome). ~900 processes, all isolated (lesson 6). *(Windows: Task Manager.)*

Identifying and reading files

```
$ file greet
greet: Mach-O 64-bit executable arm64
$ printf 'Mina\n' | xxd
00000000: 4d69 6e61 0a                                    Mina.
```

- `file` reports the **real type** from content, not the name (lessons 7, 8).
- `xxd` : **offset · hex · text** (lesson 2). `4d` =M, `69` =i, `6e` =n, `61` =a, `0a` =newline.
- `strings` pulls readable text from a program (lesson 8), a first analysis step.

Recap: who, what, files

- `whoami` → who you are and your powers (lesson 5).
- `ps` → what's running, with parents and resources (lesson 6).
- `file` + `xxd` → a file's real type and its raw bytes (lessons 2, 7, 8).

Next video: boot, and tying the whole course together.

Explore a Running System

Video 2 · Boot, and Tying It All Together

How Computers Work · Section 12

In this video, you will learn:

- How long since boot
- How five commands map to the whole course
- Where you go next

How long up, and how it started

```
$ uptime
22:32  up 6 days, 23:47, 1 user, load averages: 1.31 1.43 1.70
```

- **"up 6 days":** the boot chain (firmware → bootloader → kernel → init) finished almost a week ago (lesson 9).
- Every process traces back through **init (PID 1)** to that boot.
- "1 user" is you (lesson 5); load averages hint at CPU busyness (lesson 4).

The system also keeps **logs**, where investigators build a timeline (forensics).

Five commands, the whole course

Command	What you saw	Lesson
`whoami` / `id`	Who you are; privileges	5
`ps`	Running processes	6
`file`	A file's real type	7, 8
`xxd`	Raw bytes: offset, hex, text	2, 3
`uptime`	Time since boot	4, 9

This is the **real starting posture** of SOC triage, incident response, forensics, and reversing.

A computer is not magic

You've proven it to yourself: every concept, bits, encoding, CPU, OS, processes, memory, files, programs, boot, virtualization, is **real, visible, inspectable** with simple tools.

That demystified, confident view of the machine is the **true deliverable** of this course.

Where next: Linux (03) turns these into real skill; Networking (04) adds how machines talk; then the paths open, offensive, defensive, engineering.

Recap: you understand the machine

- Five simple commands reveal the **whole stack** this course taught.
- It's how security work **actually begins**: who, what, which files, how it started.
- A computer is **inspectable, not magic**, that's the real deliverable.
- Carry the habit: meet a new system, **look at it**, and connect what you see to how computers work.

You've finished How Computers Work. You understand the machine. Everything ahead builds on this.